

1. You are given the number of sides on a die (num_sides), the number of dice to throw (num_dice), and a target sum (target). Develop a program that utilizes dynamic programming to solve the Dice Throw Problem.

Test Cases:

1.Simple Case:

- Number of sides: 6
- Number of dice: 2
- Target sum: 7

2.More Complex Case:

- Number of sides: 4
- Number of dice: 3
- Target sum:

10 Output Test Case

1:

Number of ways to reach sum 7: 6 Test

Case 2:

Number of ways to reach sum 10: 27

Correct Code

```
def dice_throw(num_sides, num_dice, target):  
  
    dp = [[0] * (target + 1) for _ in range(num_dice + 1)]  
  
    dp[0][0] = 1  
  
    for d in range(1, num_dice + 1):  
        for s in range(1, target + 1):  
            for face in range(1, num_sides + 1):  
                if s - face >= 0:  
                    dp[d][s] += dp[d - 1][s - face]  
  
    return dp[num_dice][target]  
  
# Test Case 1  
print("Test Case 1:")  
print("Number of ways to reach sum 7:",  
      dice_throw(6, 2, 7))  
  
# Test Case 2  
print("\nTest Case 2:")  
print("Number of ways to reach sum 10:",  
      dice_throw(4, 3, 10))
```

```
*** Test Case 1:  
    Number of ways to reach sum 7: 6  
  
    Test Case 2:  
    Number of ways to reach sum 10: 6
```

2. In a factory, there are two assembly lines, each with n stations. Each station performs a specific task and takes a certain amount of time to complete. The task must go through each station in order, and there is also a transfer time for switching from one line to another. Given the time taken at each station on both lines and the transfer time between the lines, the goal is to find the minimum time required to process a product from start to end.

Input n : Number of stations on each line. $a1[i]$: Time taken at station i on assembly line 1. $a2[i]$: Time taken at station i on assembly line 2. $t1[i]$: Transfer time from assembly line 1 to assembly line 2 after station i . $t2[i]$: Transfer time from assembly line 2 to assembly line 1 after station i . $e1$: Entry time to assembly line 1. $e2$: Entry time to assembly line 2. $x1$: Exit time from assembly line 1. $x2$: Exit time from assembly line 2.

Output

The minimum time required to process the product.

Correct Code

```
[28]
✓ Os  def assembly_line(a1, a2, t1, t2, e1, e2, x1, x2, n):

    f1 = [0] * n
    f2 = [0] * n

    f1[0] = e1 + a1[0]
    f2[0] = e2 + a2[0]

    for i in range(1, n):
        f1[i] = min(f1[i-1] + a1[i],
                    f2[i-1] + t2[i-1] + a1[i])

        f2[i] = min(f2[i-1] + a2[i],
                    f1[i-1] + t1[i-1] + a2[i])

    answer = min(f1[n-1] + x1,
                 f2[n-1] + x2)

    return answer

n = 4

a1 = [4, 5, 3, 2]
a2 = [2, 10, 1, 4]

t1 = [7, 4, 5]
t2 = [9, 2, 8]

e1 = 10
e2 = 12

x1 = 18
x2 = 7

print("Minimum Time =", assembly_line(a1, a2, t1, t2, e1, e2, x1, x2, n))

▼ ... Minimum Time = 35
```

3. An automotive company has three assembly lines (Line 1, Line 2, Line 3) to produce different car models. Each line has a series of stations, and each station takes a certain amount of time to complete its task. Additionally, there are transfer times between lines, and certain dependencies must be respected due to the sequential nature of some tasks. Your goal is to minimize the total production time by determining the optimal scheduling of tasks across these lines, considering the transfer times and dependencies. Number of stations: 3

- Station times:
- Line 1: [5, 9, 3]
- Line 2: [6, 8, 4]
- Line 3: [7, 6, 5]
- Transfer times:

[[0, 2, 3],
[2, 0, 4],
[3, 4, 0]]

Dependencies: [(0, 1), (1, 2)] (i.e., the output of the first station is needed for the second, and the second for the third, regardless of the line).

Correct Code

```
[29]  
✓ Os  def min_production_time():  
  
    stations = [  
        [5, 9, 3], # Line 1  
        [6, 8, 4], # Line 2  
        [7, 6, 5] # Line 3  
    ]  
  
    transfer = [  
        [0, 2, 3],  
        [2, 0, 4],  
        [3, 4, 0]  
    ]  
  
    n = 3  
    lines = 3  
  
    dp = [[0] * n for _ in range(lines)]  
  
    for i in range(lines):  
        dp[i][0] = stations[i][0]  
  
    for j in range(1, n):  
        for i in range(lines):  
            dp[i][j] = float('inf')  
            for k in range(lines):  
                cost = dp[k][j-1]  
                if k != i:  
                    cost += transfer[k][i]  
                cost += stations[i][j]  
                dp[i][j] = min(dp[i][j], cost)  
  
    answer = min(dp[i][n-1] for i in range(lines))  
    print("Minimum Production Time =", answer)  
  
    min_production_time()  
  
✓ ... Minimum Production Time = 17
```

4. Write a c program to find the minimum path distance by using matrix form.

Test

Cases: 1)

{0,10,15,20}

{10,0,35,25}

{15,35,0,30}

{20,25,30,0}

Output: 80

2)

{0,10,10,10}

{10,0,10,10}

{10,10,0,10}

{10,10,10,0}

Output: 40

3)

{0,1,2,3}

{1,0,4,5}

{2,4,0,6}

{3,5,6,0}

Output: 12

Correct Code

```
[31]
✓ Os  def tsp(graph, visited, curr, count, cost):

    if count == len(graph):
        return cost + graph[curr][0]

    minimum = float('inf')

    for i in range(len(graph)):
        if not visited[i]:
            visited[i] = True

            minimum = min(
                minimum,
                tsp(graph, visited, i,
                    count + 1,
                    cost + graph[curr][i])
            )

            visited[i] = False

    return minimum

graph = [
    [0,10,15,20],
    [10,0,35,25],
    [15,35,0,30],
    [20,25,30,0]
]

n = len(graph)
visited = [False] * n
visited[0] = True

print("Minimum Path Distance =", tsp(graph, visited, 0, 1, 0))

... Minimum Path Distance = 80
```

5. Assume you are solving the Traveling Salesperson Problem for 4 cities (A, B, C, D) with known distances between each pair of cities. Now, you need to add a fifth city (E) to the problem.

Test Cases

1. Symmetric Distances

- Description: All distances are symmetric (distance from A to B is the same as B to A).

Distances:

A-B: 10, A-C: 15, A-D: 20, A-E: 25 B-C: 35, B-D: 25, B-E: 30 C-D: 30, C-E: 20 D-E: 15

Expected Output: The shortest route and its total distance. For example, A -> B -> D -> E -> C -> A might be the shortest route depending on the given distances.

Correct Code

```
from itertools import permutations

cities = ['A', 'B', 'C', 'D', 'E']

dist = {
    ('A', 'B'):10, ('A', 'C'):15, ('A', 'D'):20, ('A', 'E'):25,
    ('B', 'C'):35, ('B', 'D'):25, ('B', 'E'):30,
    ('C', 'D'):30, ('C', 'E'):20,
    ('D', 'E'):15
}

def distance(a, b):
    if (a, b) in dist:
        return dist[(a, b)]
    return dist[(b, a)]

start = 'A'
other = ['B', 'C', 'D', 'E']

min_cost = float('inf')
best_path = None

for perm in permutations(other):
    path = [start] + list(perm) + [start]

    cost = 0

    for i in range(len(path)-1):
        cost += distance(path[i], path[i+1])

    if cost < min_cost:
        min_cost = cost
        best_path = path

print("Shortest Route:", " -> ".join(best_path))
print("Total Distance:", min_cost)
```

```
*** Shortest Route: A -> B -> D -> E -> C -> A
    Total Distance: 85
```

6. Given a string *s*, return the longest palindromic substring in *S*.

Example 1:

Input: *s* = "babad"

Output: "bab" Explanation: "aba" is also a valid answer.

Example 2:

Input: *s* = "cbbd" Output:

"bb"

Constraints: • $1 \leq s.length \leq 1000$ • *s* consist of only digits and English letters.

Correct Code

```
def longestPalindrome(s):  
    result = ""  
    for i in range(len(s)):  
        # Odd length palindrome  
        l = r = i  
        while l >= 0 and r < len(s) and s[l] == s[r]:  
            if (r - l + 1) > len(result):  
                result = s[l:r+1]  
            l -= 1  
            r += 1  
        # Even length palindrome  
        l, r = i, i + 1  
        while l >= 0 and r < len(s) and s[l] == s[r]:  
            if (r - l + 1) > len(result):  
                result = s[l:r+1]  
            l -= 1  
            r += 1  
    return result  
  
s = "babad"  
print("Longest Palindrome:", longestPalindrome(s))  
*** Longest Palindrome: bab
```

7. Given a string *s*, find the length of the longest substring without repeating characters.

Example 1: Input: *s* = "abcabcbb" Output: 3

Explanation: The answer is "abc", with the length of 3.

Example 2: Input: *s* = "bbbbbb" Output: 1

Explanation: The answer is "b", with the length of 1.

Example 3: Input: s = "pwwkew" Output: 3

Explanation: The answer is "wke", with the length of 3.

Notice that the answer must be a substring, "pwke" is a subsequence and not a substring.

Constraints: • $0 \leq s.length \leq 5 * 10^4$ • s consists of English letters, digits, symbols and spaces.

Correct Code

```
[34] ✓ Os def lengthOfLongestSubstring(s):  
    seen = set()  
    left = 0  
    max_len = 0  
  
    for right in range(len(s)):  
        while s[right] in seen:  
            seen.remove(s[left])  
            left += 1  
  
        seen.add(s[right])  
        max_len = max(max_len, right - left + 1)  
  
    return max_len  
  
s = "abcabcbb"  
print("Output:", lengthOfLongestSubstring(s))  
  
*** Output: 3
```

8. Given a string s and a dictionary of strings wordDict, return true if s can be segmented into a space-separated sequence of one or more dictionary words.

Note that the same word in the dictionary may be reused multiple times in the segmentation.

Example 1:

Input: s = "leetcode", wordDict = ["leet", "code"]

Output: true

Explanation: Return true because "leetcode" can be segmented as "leet code".

Example 2:

Input: s = "applepenapple", wordDict = ["apple", "pen"]

Output: true

Explanation: Return true because "applepenapple" can be segmented as "apple pen apple".

Note that you are allowed to reuse a dictionary word.

Example 3:

Input: s = "catsandog", wordDict = ["cats", "dog", "sand", "and", "cat"] Output:

false

Correct Code

```
[35]
✓ Os
def wordBreak(s, wordDict):

    dp = [False] * (len(s) + 1)
    dp[0] = True

    for i in range(1, len(s) + 1):
        for j in range(i):
            if dp[j] and s[j:i] in wordDict:
                dp[i] = True
                break

    return dp[len(s)]

s = "leetcode"
wordDict = ["leet", "code"]

print(wordBreak(s, wordDict))

... True
```

9. Given an input string and a dictionary of words, find out if the input string can be segmented into a space-separated sequence of dictionary words. Consider the following dictionary { i, like, sam, sung, samsung, mobile, ice, cream, icecream, man, go, mango} Input: ilike
Output: Yes
The string can be segmented as "i like".
Input: ilikesamsung
Output: Yes The string can be segmented as "i like samsung" or "i like sam sung".

Correct Code

```
[35]
✓ Os
def wordBreak(s, wordDict):

    dp = [False] * (len(s) + 1)
    dp[0] = True

    for i in range(1, len(s) + 1):
        for j in range(i):
            if dp[j] and s[j:i] in wordDict:
                dp[i] = True
                break

    return dp[len(s)]

s = "leetcode"
wordDict = ["leet", "code"]

print(wordBreak(s, wordDict))

... True
```


10. Given an array of strings `words` and a width `maxWidth`, format the text such that each line has exactly `maxWidth` characters and is fully (left and right) justified. You should pack your words in a greedy approach; that is, pack as many words as you can in each line. Pad extra spaces ' ' when necessary so that each line has exactly `maxWidth` characters. Extra spaces between words should be distributed as evenly as possible. If the number of spaces on a line does not divide evenly between words, the empty slots on the left will be assigned more spaces than the slots on the right. For the last line of text, it should be left-justified, and no extra space is inserted between words. A word is defined as a character sequence consisting of non-space characters only. Each word's length is guaranteed to be greater than 0 and not exceed `maxWidth`. The input array `words` contains at least one word.

Example 1:

Input: `words = ["This", "is", "an", "example", "of", "text", "justification."], maxWidth = 16`
Output:

```
[ "This  is  an",  
  "example of text",  
  "justification. " ]
```

Example 2:

Input: `words = ["What","must","be","acknowledgment","shall","be"], maxWidth = 16`
Output:

```
[  
  "What must be",  
  "acknowledgment ",  
  "shall be      "  
]
```

Explanation: Note that the last line is "shall be " instead of "shall be", because the last line must be left-justified instead of fully-justified.

Note that the second line is also left-justified because it contains only one word.

Correct Code

```

[36]
✓ Os ▶ def fullJustify(words, maxWidth):

    result = []
    i = 0

    while i < len(words):

        line = []
        length = 0

        while i < len(words) and length + len(words[i]) + len(line) <= maxWidth:
            line.append(words[i])
            length += len(words[i])
            i += 1

        spaces = maxWidth - length

        if i == len(words) or len(line) == 1:
            row = " ".join(line)
            row += " " * (maxWidth - len(row))

        else:
            gaps = len(line) - 1
            even = spaces // gaps
            extra = spaces % gaps

            row = ""

            for j in range(gaps):
                row += line[j]
                row += " " * (even + (1 if j < extra else 0))

            row += line[-1]

        result.append(row)

    return result

words = ["This", "is", "an", "example", "of", "text", "justification."]
maxWidth = 16

for line in fullJustify(words, maxWidth):
    print(f'"{line}"')

```

▼ ... "This is an"
 "example of text"
 "justification. "

- Design a special dictionary that searches the words in it by a prefix and a suffix. Implement the WordFilter class: WordFilter(string[] words) Initializes the object with the words in the

dictionary.f(string pref, string suff) Returns the index of the word in the dictionary, which has the prefix pref and the suffix suff. If there is more than one valid index, return the largest of them. If there is no such word in the dictionary, return -1.

Example 1:

Input

```
["WordFilter", "f"]
```

```
[[["apple"]], ["a", "e"]]
```

Output

```
[null, 0]
```

Explanation

WordFilter wordFilter = new WordFilter(["apple"]); wordFilter.f("a", "e"); // return 0, because the word at index 0 has prefix = "a" and suffix = "e".

Correct Code

```
 class WordFilter:

    def __init__(self, words):
        self.words = words

    def f(self, pref, suff):

        for i in range(len(self.words)-1, -1, -1):

            word = self.words[i]

            if word.startswith(pref) and word.endswith(suff):
                return i

        return -1

wordFilter = WordFilter(["apple"])

print(wordFilter.f("a", "e"))
```

... 0